GPT-5.5 Plan: Minimal-Endpoint Multi-Objective Cryomicroneedle Optimization

CryoMN ML Project

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Abstract

This document specifies the next-generation implementation plan for cryomicroneedle (CryoMN) formulation optimization. The previous repository successfully used literature transfer, Gaussian-process surrogate modeling, Bayesian optimization, and wet-lab validation to improve mesenchymal stem-cell post-thaw viability. The next project must add mechanical robustness without turning the experimental workflow into an impractical mechanical-testing campaign. The proposed system therefore uses only two optimization objectives, one required screening gate, one secondary mechanical diagnostic, and one final qualitative insertion check. The main optimization objectives are post-thaw viability and Instron-derived axial critical load. Intact patch formation is modeled as a feasibility gate, initial stiffness is retained only as a diagnostic or tie-breaker, and thumb insertion is recorded only qualitatively.

1 Executive Summary

The new project should be implemented as a cost-aware, minimal-endpoint, multi-objective active-learning workflow. The purpose is not to discover the strongest possible frozen material in isolation, but to find CryoMN formulations that preserve MSC viability while forming intact, mechanically useful frozen microneedle patches.

The production objectives are:

- maximize `viability_percent`;
- maximize `critical_axial_load_N_per_needle`.

The required screening gate is:

- `intact_patch_formation_pass`.

The only secondary mechanical endpoint is:

- `initial_stiffness_N_per_mm_per_needle`.

The final insertion check is:

- `thumb_insertion_pass`, recorded only as a qualitative yes/no result with optional notes or an image link.

The optimizer should not estimate insertion force, insertion margin, or Young's modulus in version 1. Thumb insertion is deliberately uncontrolled and should not be converted into a quantitative mechanical target.

2 Endpoint Hierarchy

2.1 Screening Gate

`intact_patch_formation_pass` is measured before mechanical testing and is not a Pareto objective. It is a feasibility gate and classifier endpoint.

Default pass definition:

- the CryoMN patch is visibly filled after molding;
- the patch demolds and handles as an intact frozen microneedle patch;
- there is no slurry-like collapse, gross bending, patch disintegration, or obvious unfrozen liquid phase;
- at least 90 of 100 tips are visually intact after demolding and handling.

If the laboratory later changes the mold geometry, the 90/100 rule should be stored as a configurable threshold, not hard-coded into downstream modeling logic.

2.2 Primary Objectives

Endpoint	**Definition and use**
`viability_percent`	Post-thaw MSC viability, measured using the existing viability assay. This remains the biological objective transferred from the current repository.
`critical_axial_load_N_per_needle`	Maximum compressive load before buckling, fracture, sudden load drop, or force plateau during Instron axial compression, divided by the number of needles actually compressed. This is the mechanical objective.

2.3 Single Secondary Endpoint

`initial_stiffness_N_per_mm_per_needle` is calculated as the slope of the early linear force-displacement region, divided by the number of needles compressed. It should be stored, visualized, and optionally used as a diagnostic covariate or tie-breaker. It is not a Pareto objective in version 1.

2.4 Final Qualitative Check

`thumb_insertion_pass` is recorded only for finalists. It is a qualitative yes/no result, optionally accompanied by free-text notes or an image link. Because the applied thumb force is uncontrolled and unquantifiable, this endpoint must not be used to calculate insertion force, insertion margin, or a continuous insertion objective.

3 Instron 5942 Mechanical Protocol

The Instron 5942 is a 0.5 kN single-column test system in the 5900 series. The relevant capability for this project is controlled force-displacement testing in compression using Bluehill-compatible load and extension acquisition. The CryoMN mechanical assay should therefore be a low-temperature axial compression test against a rigid platen or equivalent rigid contact surface.

The mechanical test should export raw force-displacement data for every specimen. The importer will compute only:

- `critical_axial_load_N_per_needle`;
- `initial_stiffness_N_per_mm_per_needle`.

The raw file and observation record must include the following metadata:

- formulation identifier;
- experiment date;
- test temperature or cold-stage condition;
- number of needles compressed;
- crosshead speed;
- fixture/platen description;
- whether the curve had a clear peak, sudden drop, plateau, or no clear failure;
- operator notes.

Young's modulus should not be optimized in version 1. For frozen pyramidal CryoMN arrays, apparent modulus would be confounded by patch base deformation, fixture compliance, nonuniform tip contact, alignment, temperature drift, and thawing. The more defensible version-1 measurements are peak axial load and early force-displacement stiffness.

4 Experimental Policy

Each active-learning round should assume tight wet-lab capacity:

- 12 viability screens;
- intact patch formation screening for all 12 if enough molded material exists;
- 3–4 Instron mechanical tests;
- thumb insertion only for finalists.

Mechanical testing should be selected from candidates with `intact_patch_formation_pass = true`. One novelty or coverage exception per round is allowed if it is scientifically useful, but the exception must be flagged in the batch metadata so later analysis does not treat it as an ordinary mechanical candidate.

5 Constraints and Penalties

5.1 Ingredient Count

Ingredient count is a soft penalty, not a hard cutoff. Candidates with more than 8 active ingredients may be exported if the acquisition value justifies them, but they must be penalized and flagged.

For candidate x, let $n_{\text{active}}(x)$ be the number of active ingredients after applying the existing practical trace floor. The ingredient-count penalty should be:

$$P_{\text{ingredients}}(x) = \lambda_{\text{ingredients}} \max(0, n_{\text{active}}(x) - 8).$$

The implementation should store both the raw active count and the penalty value.

5.2 Single-Ingredient 500 mM Rule

The 500 mM rule applies to each single molar ingredient independently. It is not a total-solute-concentration penalty.

For each molar feature x_j, the concentration penalty should be:

$$P_{500\text{mM}}(x) = \lambda_{500\text{mM}} \sum_{j \in \mathcal{M}} \max(0, x_j - 0.5)^2,$$

where $\mathcal{M}$ is the set of molar-valued ingredient features and concentrations are represented in molar units. Percentage-valued ingredients should use separate registry-defined caps.

The candidate generator may still emit a candidate with one ingredient above 0.5 M if the acquisition value is compelling, but the row must be flagged as `single_ingredient_gt_500mM = true`.

5.3 Culture Media Exclusion

Culture media, basal media, salts, and buffers remain excluded as formulation variables. They may be recorded as fixed protocol context, but they must not appear as optimizable features.

6 Ingredient Registry

The next implementation should introduce a central formulation registry. This registry must define canonical names, feature names, units, synonyms, molecular weights or conversion rules, optimization bounds, percentage caps, and whether the 500 mM penalty applies.

The following ingredients must be added or activated:

Ingredient	Status	Implementation note
taurine	parsed but inactive	Activate as `taurine_M` if retained by registry rules.
sericin	parsed but inactive	Retain as percentage-valued `sericin_pct`.
raffinose	already present	Keep as `raffinose_M`.
methylcellulose	already present	Keep as `methylcellulose_pct`.
myo-inositol	new	Add as molar-valued `myo_inositol_M`.
4-methoxyphenyl beta-D-glucopyranoside	new	Add as molar-valued `methoxyphenyl_glucopyranoside_M`.
trehalose	already present	Keep as `trehalose_M`.

The registry should be the single source of truth for feature generation. Parser, candidate generator, penalty engine, batch exporter, and report code should all consume the same registry.

Current acquisition restrictions. The legacy literature import should read `data/processed/parsed_formul` but `cooh_pll`, `ficoll`, and `hes` are excluded from the active v2 feature set because they cannot currently be acquired. Separately, ingredients that are valid scientific candidates but temporarily unavailable in the lab should remain active registry features and be excluded only at selection time through `config_v2/availability.yaml`. This temporary list currently contains `taurine_M`, `sericin_pct`, `raffinose_M`, `methylcellulose_pct`, `myo_inositol_M`, `methoxyphenyl_beta_d_glucopyranoside_M`, and `trehalose_M`.

7 Data Transfer Policy

The current repository transfers into the new project as viability evidence only. The existing literature rows and 106 CryoMN wet-lab rows should seed the viability model, but no mechanical labels should be inferred from legacy data.

Recommended source assumptions:

Source	Default noise	Use
Literature viability	50 percentage points	Weak viability prior; 10× legacy wet-lab noise.
Legacy CryoMN wet-lab viability	5 percentage points	Trusted viability correction signal.
New viability	1 percentage point by default	Active biological observations; one fifth of legacy wet-lab noise unless overridden.
Intact patch formation	classifier likelihood	Feasibility model, not objective.
Instron critical load and stiffness	maximum of replicate SD, 15% of mean, and instrument floor	Mechanical objective and diagnostic.

8 Algorithm

8.1 Cold Start

Mechanical data start at zero, so the first phase should not rely immediately on a full hypervolume acquisition function. Instead, use k-center or greedy diversity selection among candidates that are predicted to be viable and likely to pass intact patch formation. Continue this cold-start phase until at least 8 mechanical observations have been collected.

Cold-start mechanical selection should:

- prefer predicted intact-patch pass candidates;
- cover chemically distinct regions near promising viability predictions;
- include one optional novelty/coverage exception per round;
- avoid duplicate or near-duplicate formulations already tested mechanically.

8.2 Main Acquisition

After mechanical cold start, use noisy hypervolume-based multi-objective Bayesian optimization. The production acquisition should be `qLogNEHVI`. The two objectives used for hypervolume are normalized:

- normalized `viability_percent`;
- normalized `critical_axial_load_N_per_needle`.

`intact_patch_formation_pass` should enter as a feasibility classifier and acquisition penalty. `initial_stiffness_N_per_mm_per_needle` should not enter the hypervolume objective set. It can be used only as a diagnostic, covariate, or tie-breaker.

Evaluation baselines should include:

- random or diversity sampling;
- `qLogNParEGO`;
- known-noise or variance-aware `qLogNEHVI` variants.

9 Computational Architecture

9.1 Directory Layout

The next project should add new numbered workflow stages without breaking the legacy viability pipeline:

- `src/08_multi_objective/01_build_database/build_database.py`: build the v2 database from legacy viability-only evidence;
- `src/08_multi_objective/02_select_candidates/select_candidates.py`: score a candidate pool and export the next wet-lab slate;
- `src/08_multi_objective/03_record_results/update_from_results.py`: ingest filled wet-lab results and technical replicates;
- `src/08_multi_objective/03_record_results/import_instron.py`: parse Bluehill/Instron CSV files into the round CSV;
- `src/08_multi_objective/04_visualization/visualize.py`: generate v2 diagnostics and candidate plots;
- `src/08_multi_objective/helper/`: non-executable shared model, registry, importer, penalty, acquisition, and selection code.

9.2 Decoupled Tables

The new data layer should avoid forcing every endpoint into one wide validation CSV. Use two persistent model-state tables and two selector-output CSV files:

Table	**Purpose**
`formulations`	One row per canonical formulation vector, including formulation ID, ingredient columns, active ingredient count, 500 mM flags, percentage cap flags, and human-readable recipe.
`observations`	One row per measured endpoint per formulation. Required columns include formulation ID, endpoint name, value, uncertainty, replicate count, assay type, source, date, and notes.

total_candidate_pool.csv	Full generated and scored candidate pool after temporary availability filtering. This file is useful for audit and troubleshooting, but it is not wet-lab input and is not persistent model state.
next_round_candidates.csv	Per-round selector output containing candidate identity, formulation values, predictions, feasibility diagnostics, mechanical-test recommendations, and blank wet-lab result columns. This file is not persistent model state; stage 03 folds filled results back into `observations`.

Required observation endpoint names:

- `viability_percent`;
- `intact_patch_formation_pass`;
- `critical_axial_load_N_per_needle`;
- `initial_stiffness_N_per_mm_per_needle`;
- `thumb_insertion_pass`.

10 Acceptance Tests

The implementation should include tests for the following scenarios:

- intact-patch gate ingestion and classifier training;
- per-ingredient 500 mM penalties;
- ingredient-count soft penalties above 8 active ingredients;
- culture media and basal buffer exclusion;
- new ingredient registry entries and synonyms;
- Instron CSV parsing for peak load and initial stiffness;
- cold-start selection restricted to predicted intact-patch-pass candidates;
- `qLogNEHVI` candidate generation after seeded formation and mechanical data;
- LaTeX compilation of this planning document.

11 Implementation Order

1. Add the registry and migrate parser/candidate code to consume it.
2. Add decoupled formulation, observation, and batch table helpers.

3. Implement the Instron importer and synthetic force-displacement test fixtures.
4. Import legacy literature and CryoMN viability data as viability-only observations.
5. Implement formation classifier training.
6. Implement mechanical cold-start selection.
7. Implement `qLogNEHVI` optimization once enough mechanical data exist.
8. Add Pareto-front evaluation, normalized hypervolume, IGD, and diagnostic plots.
9. Run regression tests to confirm the legacy single-objective viability pipeline remains usable.

12 Evidence Base

The plan is grounded in four evidence streams. First, the current repository shows that literature-only viability transfer is useful but insufficient without CryoMN-specific wet-lab correction. Second, the Emerson et al. CPA-cocktail study supports noisy hypervolume-based multi-objective Bayesian optimization over random search or purely scalarized baselines for limited-experiment formulation discovery. Third, Instron documentation confirms that the 5942/5900-series frame is appropriate for controlled force-displacement testing in compression with Bluehill-compatible acquisition. Fourth, microneedle mechanical-characterization literature supports extracting buckling or fracture load and stiffness from load-displacement curves, while warning that speed, alignment, base deformation, and number of compressed needles affect measured values.

References

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